

**Problem 1:** The temperature of one of the two heated black bodies  $T_1 = 2500\text{ K}$ . Find the temperature of the other body if the wavelength corresponding to its maximum emissive capacity exceeds by  $\Delta\lambda = 0.5\text{ }\mu\text{m}$  the wavelength corresponding to maximum emissive capacity of the first black body.

**Problem 2:** Spectral composition of solar radiation is same as that of the black body whose maximum emission corresponds to the wavelength  $\lambda = 0.48\text{ }\mu\text{m}$ . Find the mass lost by the sun every second due to this radiation. Evaluate the time interval during which the mass of the sun diminishes by 1%.

**Problem 3:** Using Planck radiation formula derive the approximate expression for space spectral density  $u(\omega, T)$  of the radiation.

- (i) In the range where  $\hbar\omega \ll k_B T$  ( Rayleigh Jeans Formula)
- (ii) In the range where  $\hbar\omega \gg k_B T$

**Problem 4:** Transform Planck formula for space spectral density  $u(\omega, T)$  of radiation from the variable  $\omega$  to the variable  $\nu$  and wavelength  $\lambda$ .

**Problem 5:** Using Wein's relation demonstrate that: (a) the most probable radiation frequency  $\omega_{pr} \propto T$  (b) maximum spectral density of the thermal radiation  $u(\omega, T) \propto T^3$  (c) Radiosity  $M \propto T^4$

**Problem 6:** Using Planck formula find the power radiated by unit area of a black body within a narrow wavelength interval  $\Delta\lambda = 1.0\text{ nm}$  close to the maximum of spectral radiation density at temperature  $T = 3000\text{ K}$  of the body.

**Problem 7:** Making use of Planck radiation formula derive the expressions determining number of photons per  $1\text{ cm}^3$  of a cavity at temperature  $T$  in spectral intervals  $\omega$  and  $(\omega + d\omega)$  and  $\lambda$  and  $(\lambda + d\lambda)$

**Problem 8:** Find the photoelectric threshold for Zinc and the maximum velocity of the photo electrons liberated from its surface by electromagnetic radiation with wavelength  $250\text{ nm}$ . ( Ans :  $332\text{ nm}$  and  $6.6 \times 10^5\text{ m/s}$ )

**Problem 9:** Illuminating the surface of the certain metal alternately with light wavelengths  $\lambda_1 = 0.35\text{ }\mu\text{m}$  and  $\lambda_2 = 0.54\text{ }\mu\text{m}$ . It was found that the maximum velocity of photoelectrons differ by a factor of  $\eta = 2$ . Find the work function of that metal. (Ans :  $1.9\text{ eV}$ )

**Problem 10:** Find the maximum kinetic energy and velocity of photo electrons liberated from the surface of lithium by electromagnetic radiation whose electric components varies with time  $E = a(1 + \cos(\omega t))\cos(\omega_0 t)$ , where  $a$  is a constant and  $\omega = 6 \times 10^{14}\text{ sec}^{-1}$  and  $\omega_0 = 3.6 \times 10^{15}\text{ sec}^{-1}$  ( Ans :  $0.38\text{ eV}$ )

**Problem 11:** Using the conservation laws, demonstrate that a free electron cannot absorb a photon completely?

**Problem 12:** Explain the following features of Compton scattering (Compton Effect) of light by matter.

- (i) The increase in wavelength  $\Delta\lambda$  is independent of the nature of the scattering substance.
- (ii) The intensity of the displaced components of the scattered light grows with the increasing angle of scattering and with the diminishing atomic number of the substance.
- (iii) The presence of non-displaced components of the scattered radiation.

**Problem 13:** A photon with energy  $150\text{ keV}$  is scattered at an angle  $120^\circ$  by a stationary free electron find the energy of the scattered photon. ( Ans :  $0.144\text{ MeV}$ )

**Problem 14:** A photon with momentum  $1.02\text{ MeV/c}$ , where  $c$  is velocity of light, is scattered by a stationary free electron and left with momentum  $0.255\text{ MeV/c}$ . Find the scattering angle of the photon? ( Ans :  $120^\circ$ )